Simone D'Onofrio

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EDUCATION

Ph.D., Physics **2022 – 2026**
ICE–CSIC / Universitat Autònoma de Barcelona *Barcelona, Spain*

Thesis: **Generalized Entropy and $f(R)$ Gravity: Cosmological Constraints from Inflation to Dark Energy** *Submitted, defense expected October 2026*
Supervisor: **Sergei D. Odintsov**

M.S., Theoretical Physics - 110L/110 **2019 – 2022**
Università La Sapienza *Rome, Italy*

Thesis: **Classical metrics from quantum amplitudes. The Reissner-Nordström-Tangherlini metric from the graviton emission of charged scalars.** (Link)
Supervisor: **Fabio Riccioni**

B.A., Physics **2016 – 2019**
Università La Sapienza *Rome, Italy*

Thesis: **Gravitational Love numbers**
Supervisor: **Andrea Maselli**

SKILLS

- **Expertise:** Theoretical and Observational Cosmology, Modified Gravity Theories, Gravitational Waves, Compact Objects, Tidal deformability, Gravity from Scattering Amplitudes
- **Computer:** Deep knowledge of C++, Python, Mathematica (including xAct package). Ability to use computing clusters and parallelize programs.
- **Languages:** English (fluent), Spanish (fluent), Italian (native) and Catalan (basic).

PUBLICATIONS

[1] Ke Chen, Simone D'Onofrio, Sergei D. Odintsov, and Tanmoy Paul. "Holographic dark energy from generalized entropies and extended infrared cut-offs". In: (Oct. 2026). arXiv: 2610.11616 [gr-qc].

[2] Simone D'Onofrio, Sayak Datta, and Andrea Maselli. "Axial tidal Love numbers of black holes in matter environments". In: *Phys. Rev. D* 114.8 (2026), p. 084024. DOI: 10.1103/7qcf-c8jv. arXiv: 2605.02633 [gr-qc].

[3] Simone D'Onofrio, Dong Ha Lee, Eleonora Di Valentino, and Juan García-Bellido. "General Relativistic Entropic Acceleration at the perturbation level: a CLASS implementation and first Boltzmann-code constraints". In: (July 2026). arXiv: 2607.25841 [gr-qc].

[4] Simone D'Onofrio, Sergei Odintsov, and Tanmoy Paul. "Fitting NANOGrav 15-year data and ACT data with modified inflation in entropic cosmology". In: *Phys. Rev. D* 113.4 (2026), p. 043527. DOI: 10.1103/ktt2-lm19. arXiv: 2510.20484 [gr-qc].

[5] Simone D'Onofrio, Sergei Odintsov, and Tiziano Schiavone. "Exponential f(R) cosmology with massive neutrinos as a dynamical dark energy framework". In: *Annals Phys.* 494 (2026), p. 170621. DOI: 10.1016/j.aop.2026.170621. arXiv: 2511.06924 [gr-qc].

[6] Simone D'Onofrio. "I-Love-Q relations for neutron stars with dark energy". In: *Eur. Phys. J. C* 85.6 (2025), p. 672. DOI: 10.1140/epjc/s10052-025-14397-2. arXiv: 2506.01889 [gr-qc].

[7] Sergei D. Odintsov, Simone D'Onofrio, and Tanmoy Paul. "Generalized entropic dark energy with spatial curvature". In: *Phys. Dark Univ.* 48 (2025), p. 101920. DOI: 10.1016/j.dark.2025.101920. arXiv: 2504.03470 [gr-qc].

[8] Sergei D. Odintsov, Simone D'Onofrio, and Tanmoy Paul. "Entropic Inflation in Presence of Scalar Field". In: *Universe* 10.1 (2024), p. 4. DOI: 10.3390/universe10010004. arXiv: 2312.13587 [gr-qc].

[9] Sergei D. Odintsov, Simone D'Onofrio, and Tanmoy Paul. "Primordial gravitational waves in horizon cosmology and constraints on entropic parameters". In: *Phys. Rev. D* 110.4 (2024), p. 043539. DOI: 10.1103/PhysRevD.110.043539. arXiv: 2407.05855 [gr-qc].

[10] Simone D'Onofrio. "Holographic description of F(R) gravity coupled with axion dark matter". In: *Int. J. Geom. Meth. Mod. Phys.* 20.13 (2023), p. 2350224. DOI: 10.1142/S0219887823502249. arXiv: 2306.16743 [gr-qc].

[11] Sergei D. Odintsov, Simone D'Onofrio, and Tanmoy Paul. "Holographic realization from inflation to reheating in generalized entropic cosmology". In: *Phys. Dark Univ.* 42 (2023), p. 101277. DOI: 10.1016/j.dark.2023.101277. arXiv: 2306.15225 [gr-qc].

[12] Simone D'Onofrio, Federica Fragomeno, Claudio Gambino, and Fabio Riccioni. "The Reissner-Nordström-Tangherlini solution from scattering amplitudes of charged scalars". In: *JHEP* 09 (2022), p. 013. DOI: 10.1007/JHEP09(2022)013. arXiv: 2207.05841 [hep-th].

[13] Simone D'Onofrio. "Classical metrics from quantum amplitudes: The Reissner-Nordström-Tangherlini metric from the graviton emission of charged scalars". MA thesis. Rome U., 2021.

GRANTS AND AWARDS

FPI Predoctoral Fellowship, Spanish Ministry of Science and Innovation **2022–2026**
Competitive four-year national PhD contract at ICE-CSIC, Barcelona (Agencia Estatal de Investigación, ref. PRE2021-098098).

Simons Strong Gravity Visit Grant **25 Jan – 5 Feb 2027**
1,100 USD. Long-Term Visit Program, Simons Collaboration on Black Holes and Strong Gravity. Supports my visit to the Niels Bohr Institute, University of Copenhagen, with Prof. Vitor Cardoso.

Short-Term Scientific Mission grant, COST Action CA21136 (CosmoVerse) **28 Sep – 9 Oct 2026**
2,000 EUR. Funded my research stay at the University of Sheffield with Prof. Eleonora Di Valentino.

CosmoVerse@Bilbao 2026 Conference Grant **15–18 Sep 2026**
1,000 EUR plus flights. COST Action CA21136 (CosmoVerse). Travel and per diem support, awarded through a competitive selection together with a contributed talk.

CosmoVerseSchool@Sofia 2026 Training School Grant **25–29 May 2026**
1,190 EUR. COST Action CA21136 (CosmoVerse). Travel and per diem support (flights included).

Travel grant from ILANCE (CNRS / University of Tokyo) **17–21 Nov 2025**
1,000 EUR. Covered the flight to Tokyo for the 2nd International Conference on the Physics of the Two Infinities.

Conference fee waivers **2025–2026**
International Conference Dark Matter and Stars (ICDMS 2026), University of Southampton; PASCOS 2026, University of Sheffield; 2nd International Conference on the Physics of the Two Infinities (2025), University of Tokyo.

CONFERENCES

CosmoVerse@Bilbao 2026 **15–18 Sep 2026**
Bilbao, Spain. 25-minute contributed talk, selected competitively. Travel grant awarded. Talk title: Dynamical dark energy from exponential f(R) gravity: cosmological tensions and neutrino mass after DESI DR2.

Dark Matter and Stars (ICDMS) **13–15 Jul 2026**
University of Southampton. 20-minute contributed talk. Fee waiver awarded. Talk title: Tidal Love Numbers of Black Holes Dressed by Dark Matter.

PASCOS 2026 **22–26 Jun 2026**
String/Black Hole parallel session. University of Sheffield. 20-minute presentation. Fee waiver awarded. Talk title: Tidal Love Numbers of Black Holes Dressed by Dark Matter.

CosmoVerseSchool@Sofia 2026 **25–29 May 2026**
Institute of Mechanics, Bulgarian Academy of Sciences, Sofia. COST Action CA21136 training school. 2-minute lightning talk. Travel grant awarded.

2nd Int. Conference on the Physics of the Two Infinities **17–21 Nov 2025**
Cosmology parallel session. University of Tokyo. 15-minute presentation. Funded by ILANCE-CNRS. Talk title: Generalized entropic cosmology.

6th BIG&C Meeting **4 Jul 2025**
ICE-CSIC, Barcelona. 20-minute presentation. Talk title: Generalized entropies as dark energy models.

Gravity 2025 Conference **27–31 Jan 2025**
YITP. Yukawa Institute for Theoretical Physics, Kyoto. Poster presentation. Talk title: Generalized entropies as dark energy models.

GSSI–JHU Meeting **16–18 Dec 2024**
Assergi, Italy. 15-minute talk. Talk title: Generalized entropic cosmology: from inflation to reheating.

SCIENTIFIC ACTIVITIES

Seminar at University of Sheffield **8 Oct 2026**
One-hour seminar (host: Eleonora Di Valentino). Talk title: Tidal deformability of compact objects and the effects of an environment.

Seminar at La Sapienza University, Rome **23 Sep 2026**
One-hour seminar (host: Paolo Pani). Talk title: Tidal deformability of compact objects and the effects of an environment.

Invited Seminar at Queen Mary University of London **8 Jul 2026**
One-hour invited seminar. Talk title: Tidal deformability of compact objects and the effects of an environment.

Invited Seminar at Keio University, Tokyo **17 Nov 2025**
20-minute invited seminar (host: Antonino Flachi). Talk title: Generalized entropic cosmology.

Invited Seminar at Università degli Studi di Cagliari **12 Nov 2025**
20-minute invited seminar (host: Riccardo Murgia). Talk title: Generalized entropic cosmology.

Refereeing activities
I have refereed more than ten papers for various journals, among which Physics of the Dark Universe and JHEAP.

VISITING EXPERIENCES

Visiting researcher at Niels Bohr Institute (upcoming) **25 Jan – 5 Feb 2027**
University of Copenhagen, with Prof. Vitor Cardoso, funded by the Simons Collaboration on Black Holes and Strong Gravity

Invited visiting student to University of Sheffield **28 Sep – 9 Oct 2026**
With Prof. Eleonora Di Valentino and funded by the STSM project of the COST Action CA21136

Visiting student to University of Sheffield **Jun–Jul 2026**
With Prof. Eleonora Di Valentino

Invited visit to Università degli Studi di Cagliari **8–14 Nov 2025**
With Prof. Riccardo Murgia, funded by host institution

Invited visit to GSSI institute **9–13 Jun 2025**
With Prof. Andrea Maselli, funded by host institution

Visiting student to GSSI institute **Nov–Dec 2024**
With Prof. Andrea Maselli

Visiting student at Università Federico II of Naples **Sep–Oct 2023**
With Prof. Mariafelicia De Laurentis

OUTREACH ACTIVITIES

(All of the following activities were performed during my PhD in Spain and all of them have been done in Spanish.)

MAGNET Project **2023–2025**
I was part of the Magnet project which consists of a Spanish campaign to link research institutes to primary schools with social problems. I organized and carried out various activities with classes of young kids about various topics of physics during all the duration of the project.

Workshop at *Researchers' night* in UAB **26 Sep 2025**
Organised an activity at the UAB University for the Researchers' night in which we presented basic concepts of General Relativity to the participants of the event.

Workshop at *Festa de la Ciencia* in Barcelona **31 May 2025**
Together with Jorge Riviero, we organised an activity at the science fair of the municipality of Barcelona named "Life and death of stars", in particular I explained the formation of black holes and their properties.

Lecture *Ciencia en el Barrio* **20 Mar 2025**
Together with Michele Lenzi, we organised a lecture with the CSIC (Spanish national research institute) to explain General Relativity and Cosmology to students of the High School "IES Miquel Tarradell" of Barcelona. The students then went to the National Fair of "Ciencia en el Barrio" in Sevilla (one of the biggest outreach events in Spain) to present what they learned from the activity.

Lecture *Ciencia en el Barrio* **14 Mar 2025**
Together with Michele Lenzi, we organised a lecture with the CSIC (Spanish national research institute) to explain General Relativity, Black Holes and Gravitational Waves to students of the High School "Institut Front Marítim" of Barcelona.

Workshop at *Researchers' night* in UAB **27 Sep 2024**
Together with Michele Lenzi, we organised an activity at the UAB University for the Researchers' night in which we presented basic concepts of General Relativity to the participants of the event.

Workshop *Ciencia en el Barrio* **9 Jun 2024**
Together with Jorge Riviero, we organised a workshop with the CSIC (Spanish national research institute) to explain the Dark Universe to the people of the city.

Workshop at *Saló de l'Ensenyament*, Barcelona's annual fair **13 Mar 2024**
Together with Jorge Riviero, we organised a workshop on Black Holes at the Barcelona's annual university fair.